# Kaleb Coleman

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## SUMMARY

Incoming M.S. student in Artificial Intelligence Engineering (Robotics) at Arizona State University with experience developing machine learning models, deep learning systems, and large-scale analytical workflows. Built projects in generative modeling, computer vision, sports analytics, and data engineering using Python, SQL, and PyTorch. Interested in machine learning, generative modeling, autonomous systems, simulation, and intelligent decision-making.

## EDUCATION

M.S., Artificial Intelligence Engineering (Robotics), Expected May 2028
Arizona State University, Mesa, AZ

B.S., Data Science; Cybersecurity Minor, May 2026
Northern Arizona University, Flagstaff, AZ

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, C, JavaScript
**ML / AI:** PyTorch, scikit-learn, XGBoost, PyGAM, PCA, clustering, regression, autoencoders, diffusion models, computer vision
**Data / Systems / Compute:** SQLite, PostgreSQL, JSON parsing, schema validation, API ingestion, Linux/Unix, Docker, Git/GitHub, HPC workflows, GPU acceleration

## PROJECTS

**Cross-Sport Shot Analysis with GAM and Shot Difficulty Index Modeling** Spring 2025 – Spring 2026

- Engineered an end-to-end pipeline to ingest, clean, validate, store, and model NBA and NHL shot data from 2014–2024, **spanning approximately 3 million rows**, converting raw JSON into structured relational tables
- Implemented schema validation, integrity checks, and idempotent upsert logic for dataset updates
- Built generalized additive models (GAMs) for shot difficulty and scoring trend analysis using spatial shot features
- Developed a Shot Difficulty Index (SDI) framework for evaluating shot quality and scoring performance
- Created visual analytics comparing shot difficulty, efficiency, and player scoring patterns
- Presented as part of undergraduate capstone research through a formal poster presentation at NAU UGRADS

**Deep Learning for Image Reconstruction & Generation (Autoencoder, PCA, Diffusion)** Spring 2026

- Built and evaluated PyTorch autoencoders across latent dimensions **(k=2–64)** using MNIST and Fashion-MNIST
- Conducted 5-fold cross-validation comparing nonlinear autoencoders against PCA baselines
- Improved reconstruction quality **(SSIM 30 → 88)** through architecture tuning and training optimization
- Implemented a DDPM-style diffusion model from scratch using a UNet-based architecture and timestep embeddings
- Built modular components for noise scheduling, training, validation, and reverse diffusion sampling
- Accelerated deep learning experimentation using GPU resources and scalable training workflows
- Structured the codebase for reproducible experimentation and scalable model development

**Automated Course Data Pipeline for AI Curriculum Discovery** Spring 2026

- Engineered a Python pipeline to scrape, extract, and aggregate **thousands of NAU course offerings** from web sources
- Processed and structured raw web data into a centralized, queryable format for institutional analysis
- Developed custom classification logic to identify AI-related coursework across diverse academic departments
- Supported the **IAAAI initiative** by streamlining AI curriculum discovery, academic planning, and course exploration

## AWARDS & HONORS

**NAU ASA DataFest 2026 — Best Visualization Award** Spring 2026

- Competed in a **48-hour** national data science competition, analyzing a large-scale, real-world dataset.
- Developed high-impact visual analytics to identify key behavioral trends and anomalies.
- Produced presentation-quality visualizations that secured the top award and a 1-year membership to the American Statistical Association (ASA).